The 204 structural rows of the companion M_{23} table

Private L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and named-model interfaces, each of the 204 subgroup-class representatives in the current Connor–Leemans M_{23} table has the structural description printed in its row. A compound label asserts the existence of the displayed normal factors, complements or nonsplit extensions; it need not characterize a unique isomorphism type.*

Source and numerical interface. The ten-page companion PDF is dated 25 November 2012. All ten native pages were directly inspected. The separately accepted robust numerical audit supplies the complete 204-class catalogue, 679 weighted maximal inclusions, and the exact source-row bijection. Those results and their computational classification premises are acquired unchanged; no ambient subgroup census is repeated here. The original and reweighted group files contain only literal permutation groups, so every proper subgroup keeps its actual 23-point embedding. This proves a current companion result, without asserting historical PDF byte identity or discharging the other larger tables.

The source's default left association is retained. In particular row 42 is $2\cdot L_2(7)$, not $2 : L_2(7)$. Baseline integers and superscripts are recorded separately: row 49 is $23 : 11$ and row 155 is the cyclic group of order 23, whereas row 188 has the elementary abelian label 2^3. The flat text string 23 is insufficient to distinguish these cases; the explicit native assertion ledger makes the distinction.

Named ambient group. The acquired driver constructs `MathieuGroup(23)`. Its clean complete receipt, all recorded input/output hashes, and actual generator equality with the numerical catalogue are checked. GAP Reference Manual 50.1-13 supplies the named constructor interface. The numerical top representative is an actual subgroup of this group with the full ambient order 10 200 960, and hence is the group itself. This acquisition pays row 1 without enumerating ten million elements anew. Its name is not inferred from an abstract order match.

Actual structural transports. The registry acquires 224 earlier accepted named and elementary models. For each proper row, the certificate specifies images of all ordered model generators. The independent reader closes the source model and verifies that the induced word map is a bijection onto the actual embedded subgroup, fixes identity and preserves every right-generator edge. Induction on words, as in the acquired MC.3 transport theorem, makes this map a homomorphism; it is therefore an isomorphism. All previously proved atom identifications retain their original authorship and exact acceptance routes.

The five new models. Four new models are split extensions:

$$2^4 : A_7, \qquad (2^3 : L_2(7)) : 2, \qquad 2^4 : L_2(7), \qquad (((2 \times 4) : 2) : 2) : 2 : S_3.$$

For each extension the reader closes actual normal and complement subgroups, checks their generator conjugates, trivial intersection and product of orders, and verifies the factor isomorphisms. The nontrivial displayed actions have noncommuting factor generators. These are explicit existence certificates, not conclusions from a structure-description heuristic.

The fifth model is $2 \cdot L_2(7)$. Its actual normal subgroup N has order two and is central. Literal cosets give a faithful quotient action, which an explicit generator-edge bijection identifies with the acquired $L_2(7)$ model. Two actual elements x, y have nonidentity commutator in N. Their images in the quotient commute. Centrality gives $[nx, my] = [x, y] \neq 1$ for all $n, m \in N$. A complement would provide commuting lifts of those commuting quotient elements, a contradiction. Thus the extension is nonsplit. This is an application of the already accepted central commuting-lifts criterion, whose original contribution by root is retained; no new general criterion is claimed here.

Finite certificate closure. The literal reader proves every new model from previously verified factors, then all 203 proper occurrence maps. Together with the named ambient join this covers all 204 rows through 213 maps and 1,848,806 generator edges. Every source order, exact name tree, embedded generator array and source-row bijection is checked against the acquired inputs. The five new models, the completed witness records and their raw execution receipts are retained. GAP discovery only proposes witnesses; correctness is established by literal permutation arithmetic.

The first bounded discovery stopped at the unsupported nonabelian quotient case in row 42. Its forty completed witnesses are retained individually, without calling that execution a complete success. A successor adds only candidate generation for the already proved central criterion and preserves those forty rows. The two baseline-23 ambiguities then produced explicit `NO_CERTIFICATE` results; after native correction only those two rows were retried. Their new witnesses pass the same independent reader. No failed candidate search is evidence of nonsplitting or of a missing subgroup.

Ordinary and optimized Python must return identical complete reports. Focused damages cover missing rows, wrong embeddings and degrees, altered source pins and names, dots changed to colons, baseline 23 changed to 2^3, malformed syntax and an unlicensed displayed constructor. The acquired extension and generator-edge controls remain explicit dependencies. This application is authored by `1044_resume`; its independent review is separate. It claims no new original, question, measurement, formalization, canonical label uniqueness or whole-L054 completion.